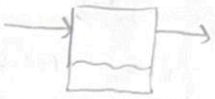
CBE 150B Cheat Sheet

Drying

A = H_2O

Absolute Humidity: $H = \frac{P_A}{P_{tot} - P_A}\left(\frac{18.02}{28.97}\right)$; relative humidity: $RH = (100)\frac{P_{H_2O}}{P^{sat}_{H_2O}}$; $\frac{H - \tilde{H}_s}{T - T_s} = \frac{-C_p}{\lambda_{H_2O}}$ [Energy needed to achieve a certain level of drying]

Enthalpy → Energy Balance



$\%H = (100)\frac{H}{H_s}$; $\tilde{H}_s = C_p(T - T_{ref}) + H\lambda_{H_2O}$; $V_H = (2.83\times10^{-3} + 4.56\times10^{-3}H)\,T$ [=] $\frac{m^3}{kg\ DA}$

$C_i = \frac{\rho}{MW}$; Humid heat: $C_p = 1.005 + 1.88H$ [=] $kg/kg\ DA \cdot K$; $V_H = (0.0252 + 0.0405H)\frac{T}{MW_{DA}}$ [=] $\frac{ft^3}{lb_m\ DA}$ ($R = F + 459.67$)

$\bar{X} = \frac{W_{total} - W_{dry}}{W_{dry}}$; $\frac{\partial \bar{X}}{\partial t} = D_L\frac{\partial^2 \bar{X}}{\partial y^2} \Rightarrow \frac{\bar{X}(t)}{\bar{X}_c} = \frac{8}{\pi^2}\exp\left(-Dt\left(\frac{\pi}{2y_1}\right)^2 t\right)$; $q = k(T - T_w)$

Rate of Drying: $R = \frac{-W_{dry}}{A}\frac{d\bar{X}}{dt}$; $R = \frac{q}{\Delta H_{vap}}$

BC: 1) $\bar{X}(y, t=0) = \bar{X}_c$

2) $\bar{X}(y = y_1, t) = 0$

3) $\left.\frac{\partial \bar{X}}{\partial y}\right|_{y=0} = 0$

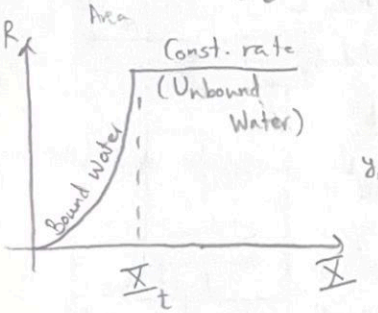
Time to dry:

1) $t = \frac{W_{dry}}{A\left(\frac{k(T - T_w)}{\lambda_{H_2O}}\right)}(X_c - X_f)$ [Constant rate] mass transfer limited

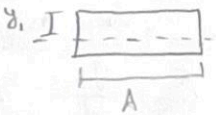
(mass transfer coefficient)

2) $t = \frac{4y_1^2}{\pi^2 D_t}\ln\left(\frac{8\bar{X}_c}{\pi^2\bar{X}_f}\right)$ [Changing rate] Diffusion limited

(D_t: Diffusivity of H_2O in slab)



Area



$h = 0.0204\,G^{0.8}$ [$J/m^2 s K$] } Parallel flow

$G = gv$ [$\frac{kg}{m^2 h}$]

$h = 1.17\,G^{0.37}$ (Perp. flow)

Thermodynamics

EB: $\sum_{out}(nh + Q + W_s) - \sum_{in}(nh + Q + W_s) = 0$

T_s = coolant / heatant temp

T_0 = Surroundings T

$\eta_{carnot} = \frac{W}{Q_{in}} = \frac{1 - T_{out}}{T_{in}}$

Entropy Balance: $\sum_{out}\left(ns + \frac{Q}{T_s}\right) - \sum_{in}\left(ns + \frac{Q}{T_s}\right) = \Delta S_{irr}$

$\Delta S_{univ} \geq 0$

Lost work

$\eta = \frac{W_{min}}{LW + W_{min}}$

Availability balance: $\sum_{in}\left[nb + Q\left(1 - \frac{T_0}{T_s}\right) + W_s\right] - \sum_{out}\left[nb + Q\left(1 - \frac{T_0}{T_s}\right) + W_s\right] = LW = T\Delta S_{irr}$

$b = h - T_0 s$ (Availability)

$W_{min} = \sum_{out} nb - \sum_{in} nb$

$G = H - Ts$

$W_{min} + W_{lost} = W = \sum_{in} Q_{in}\left(1 - \frac{T_0}{T_{in}}\right) - \sum_{out} Q_{out}\left(1 - \frac{T_0}{T_{out}}\right) + W_{s,in} - W_{s,out}$

$\phi_i y_i P = x_i \gamma_i P_i^{sat}$; $K = \frac{y_i}{x_i} = \frac{y_i\phi_L}{\phi_V} = \frac{\gamma_i P_i^{sat}}{\phi_i P} \approx \frac{P_i^{sat}}{P}$; $\alpha_{AB} = \frac{K_A}{K_B} = \frac{P_A^{sat}}{P_B^{sat}}$ → Relative volatility

($\alpha_{AB} > 1$ Easy separation)

$f_{iV} = f_{iL}$

diffuses

Gibbs

Intensive: $F = C - P + 2$

$\nu = cP + 2^{P}$ (Int. vars)

$\xi_c = P + C(P-1)$

$\Sigma x_i = 1 \quad K_i = \frac{y_i}{x_i}$

$\Sigma y_i = 1$

$F = \nu - \xi_c$

System

$\nu = \nu_{int} + (C+P+4)$

$\xi_c = \xi_{c\,int} + (C+1)$

$F = \nu - \xi_c$

$= C + 5$

All separations

Flash

$F = C + 5$

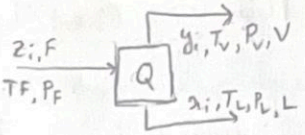
$\xi_c = 3C + 10$

$(z_i; x_i; y_i, 3T, 3P, Q, F, V, L)$

Feed defined $(C+3)$

$\therefore \xi_c = 2C + 5$

vars undefined = 2



$3C = \{ z_i, x_i, y_i \}$

$10 = \{ 3T, 3P, F, V, L, Q \}$

$T_V, P_V \Rightarrow$ Isothermal

$V/F = 0, P_L \Rightarrow$ Bubble point

$V/F = 1, P_V \Rightarrow$ Dew-point

$T_L, V/F = 0 \Rightarrow$ Bubble point pressue

$T_V, V/F = 1 \Rightarrow$ Dew-point pressue

$Q = 0, P_V \Rightarrow$ Adiabatic flash

$G, P_V \Rightarrow$ Non-adiabatic flash

$V/F, P_V \Rightarrow$ Percent vap. flash

Flash: $\sum_{i=1}^{C} \frac{z_i(1-K_i)}{1+\Psi(K_i-1)} = 0$; $\frac{\Psi-1}{\Psi} x_A + \frac{1}{\Psi} z_A = y_A$; $\Psi = \frac{V}{F}$; $P_{bubble}(\Psi=0) = \sum z_i P_i^{sat} \quad (z_i = x_i)$

(Rachford rice)

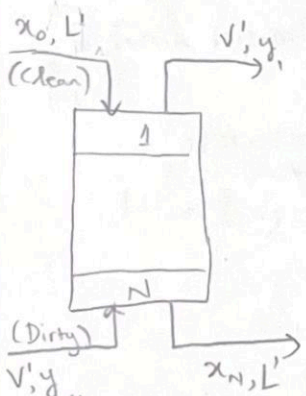
$P_{dew}(\Psi=1) = \frac{1}{\sum \left(\frac{z_i}{P_i^{sat}}\right)} \quad (z_i = y_i)$

$P_i^{sat} = \exp\left(A - \frac{B}{T+C}\right)$

Absorption & Stripping

Absorber (gas → liq), Stripper (liq → gas)

$S = 1/A$



$\phi_A = \frac{v_1}{v_{N+1}} = \frac{y_1 V'}{y_{N+1} V'}$ (fraction not removed $(0 < \phi < 1)$)

$A_e = \frac{L}{VK}$ (clean / dirty)

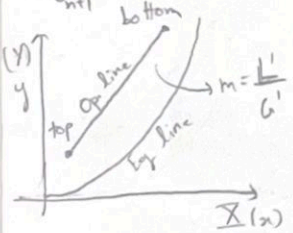
$\phi_A = \frac{A_e - 1}{A_e^{N+1} - 1}$

$\overline{X}_n = \frac{x_n}{1 - x_n}$

$\overline{Y}_n = \frac{y_N}{1 - y_n}$

Optimum $L = 1.5\, L_{min}$

Optimum $A_e = 1.4$



$\left(\frac{L'}{G'}\right)_{min} = \frac{(\overline{Y}_{n+1} - \overline{Y}_1)}{\left[\frac{\overline{Y}_{n+1}}{\overline{Y}_{n+1}(K_n - 1)} + \overline{X}_n\right] - \overline{X}_0}$

Op. line:

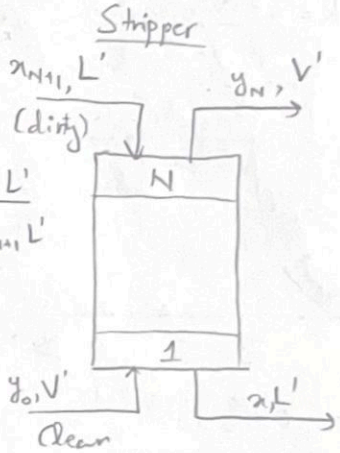
$\overline{Y}_{n+1} = \overline{X}_n \left(\frac{L'}{G'}\right) + \overline{Y}_1 - \overline{X}_0 \left(\frac{L'}{G'}\right)$

$\overline{Y}_n = \overline{X}_{n+1} \left(\frac{L'}{G'}\right) + \overline{Y}_0 - \overline{X}_1 \left(\frac{L'}{G'}\right)$

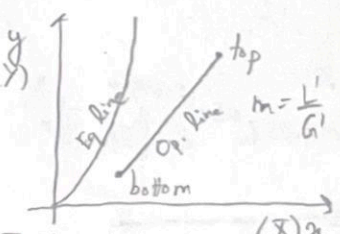
$\phi_S = \frac{l_1}{l_{N+1}} = \frac{x L'}{x_{N+1} L'}$

$S_e = \frac{VK}{L}$

$\phi_S = \frac{S_e - 1}{S_e^{N+1} - 1}$



$\left(\frac{L'}{G'}\right)_{opt} = 1.5 \left(\frac{L'}{G'}\right)_{min}$



$\overline{X}_0 L' + \overline{Y}_{n+1} G' = \overline{X}_n L' + \overline{Y}_1 G'$

Packed bed

$k_L = \frac{1}{L} \int_0^L k_x \, dz$

Length = $H_{OG} \cdot N_{OG}$

$\frac{k_x a}{C_t D_{AB}} \cdot \frac{k_L a}{D_{AB}} \quad Sh = \alpha \left(\frac{L}{\mu_L}\right)^{1-n} \left(\frac{\mu_L}{g D_{AB}}\right)^{0.5}$

$\frac{1}{\overline{K}_G C_g} = \frac{1}{k_g C_g} + \frac{K}{k_L C_L}$

$\frac{1}{\overline{K}_L a} = \frac{1}{k_L a} + \frac{1}{K k_g a}$

H_{OG}

$(y - y^*) \quad \frac{V}{\overline{K}_y a S}$

$(p - p^*) \quad \frac{V}{\overline{K}_G a P S}$

$(\overline{Y} - \overline{Y}^*) \quad \frac{V'}{\overline{K}_y a S}$

H_{OL}

$(x - x) \quad L/\overline{K}_x a S$

$(c^* - c) \quad L/(K_L a (P/M) S)$

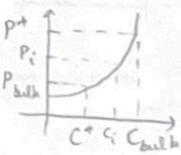
$(\overline{X} - \overline{X}) \quad L'/(\overline{K}_x a S)$

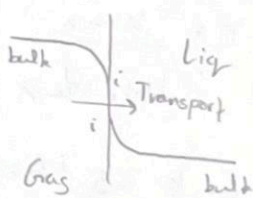
$N_A = k_G (P_{A,i} - P_A) = \overline{K}_G (P_A^* - P_A)$

$= k_L (C_A - C_{A,i}) = \overline{K}_L (C_A - C^*)$

$N_A = k_y (y_{A,i} - y_A) = \overline{K}_y (y_A^* - y_A)$

$= k_x (x_A - x_{A,i}) = \overline{K}_x (x_A - x_A^*)$



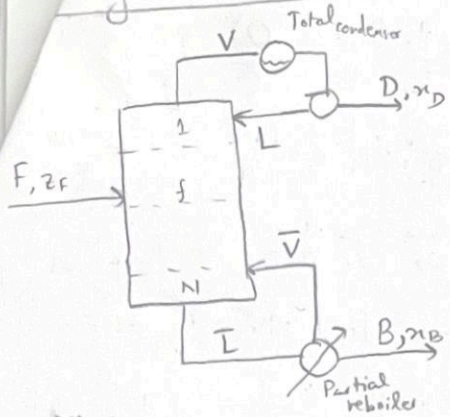


$N_{OG} = \frac{\ln\left[\left(\frac{A-1}{A}\right)\left(\frac{y_{in} - K x_{in}}{y_{out} - K x_{in}}\right) + \frac{1}{A}\right]}{\left(\frac{A-1}{A}\right)}$

diffuses

Drying ... humidity

Binary Distillation

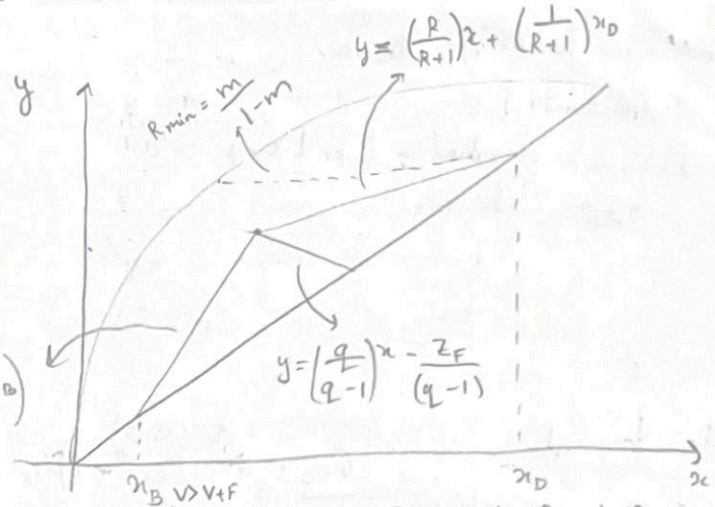


$R = \frac{L}{D}$ (Reflux Ratio)

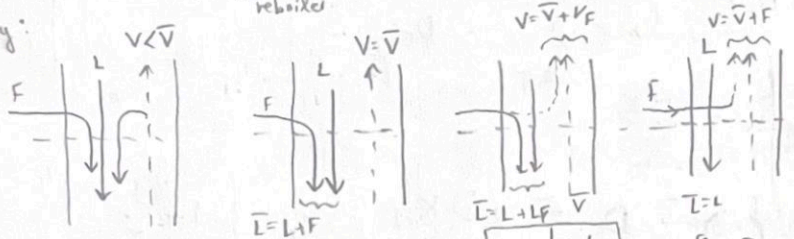
$V_B = \frac{\bar{V}}{B}$ (Reboiler Ratio)

Optimum: $R = 1.3\,R$

$y = \left(\frac{V_B+1}{V_B}\right)x - \left(\frac{1}{V_B}\right)x_B$



Quality:

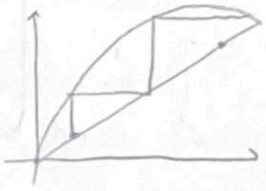


- $q > 1$: Subcooled liquid
- $q = 1$: Sat. liquid
- $q = L_F/F$: partially vaporized
- $q = 0$: Sat. vapor
- $q < 0$: superheated vapor

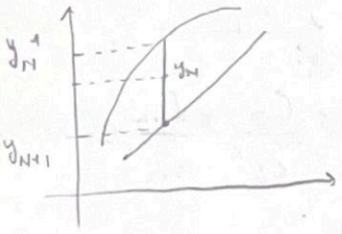
Total reboiler/reflux

$R \to \infty \Rightarrow N_{min}$

(least capital costs & highest opex)



+ Murphree efficiency: How close it reaches equilibrium



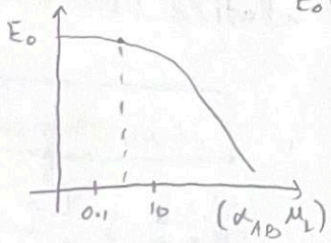
$E_M = \frac{y_N - y_{N+1}}{y_N^+ - y_{N+1}}$

McCabe Thiele Assumptions:

1) No heat loss
2) Const. P
3) Components have equal & const. latent heat
4) $c_p\Delta T$ & ΔH_{mix} are negligible
5) Binary mixture

+ Overall efficiency:

$E_o = \frac{N_{theoretical}}{N_{Actual}} = 50.3\,(\alpha_{AB}\,\mu_{Liq})^{-0.226}$

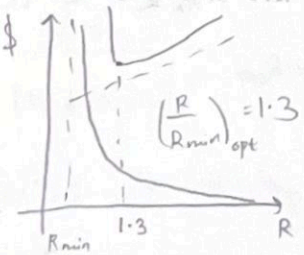


+ Energy balance on Dist:

$Fh_F + Q_R = Dh_D + Bh_B + Q_C + \cancel{Q_{loss}}^{0}$

Condenser: $Q_C = \underbrace{D(R+1)}_{V}\Delta H^{vap}$

Reboiler: $Q_R = \underbrace{BV_B}_{\bar{V}}\Delta H^{vap}$



$V = \bar{V} + V_F$ (partial vaporized)

$(L+D) = V_B \cdot B + V_F$

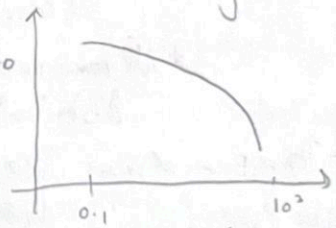
$D(R+1) = V_B \cdot B + V_F$

$Q_R = Q_R - V_F$

$Q_R = Q_R\left(1 - \frac{V_F}{Q_R}\right) = Q_R\left(1 - \frac{V_F}{D(R+1)}\right)$

V_F/F for flash

* Plate efficiency (specific to each stage)



$E_o \propto \left(K_L\, M_L\, \mu_L/\rho_L\right)^{-1}$

- K_L: Eq Const
- M_L: MW of liq
- μ_L: visc. of liq
- ρ_L: density of liq.

Assumptions for stripper / Absorber:

1) Constant L, V
2) No air dissolved in L or L evap. in air
3) Ideal gas for air

+ Stage (count the reboiler as 0)

+ Tray (count the reboiler as 1)

+ Partial Reboiler → 1st stage is R

" Condenser → Last stage is R

New formulas

+ Membrane separation:

Porous

Gas: $N_i = \frac{D_{ei}\, C_m (P_{i0} - P_{iL})}{P_{lm}}$; $C_m = \frac{\rho}{RT}$; $D_{ei} = \frac{\varepsilon}{\tau}\left[\frac{1}{\frac{1}{D_i} + \frac{1}{D_{ki}}}\right]$

Permeance ⇒ $P_{mi} = \frac{D_{ei}\, C_m}{\rho}$

Liq: $N_i = \frac{D_{ei}}{l_m}(C_{i_0} - C_{iL})$; $D_{ei} = \frac{\varepsilon D_i K_{ri}}{\tau}$; $K_{ri} = \left[1 - \frac{d_{mi}}{d_p}\right]^4$

Permeance: $\frac{\varepsilon D_i K_{ri}}{\tau}$

Non-Porous

Gas: $N_i = \frac{H_i D_i}{l_m}(P_{i0} - P_{iL})$; $H_{i0} = \frac{C_{i0}}{P_{i0}} = \frac{C_{iL}}{P_{iL}}$; $P_{mi} = H_i D_i$

Liq: $N_i = \frac{D_i}{l_m}(C_{i0} - C_{iL}) = \frac{K_i D_i}{l_m}(C_{i0}' - C_{iL}')$; $K_i = \frac{C_{i0}}{C_{i0}'} = \frac{C_{iL}}{C_{iL}'}$

$P_{mi} = K_i D_i$

+ Multidistillation: (Nmin) Frenske → (Rmin) Underwood → Gilliland (Real N, & Real R_{min})

$\alpha = \frac{y_i}{x_i}$ ↑α means it's more volatile (Lower key)

Class I
+ All components distribute

Class II
+ Only LK & HK distribute

$$R_{min} = \frac{(L_0)_{min}}{D}$$

$$\frac{(L_0)_{min}}{F} = \left(\frac{L_F}{F}\right)\left[\frac{\frac{D x_{LK,D}}{L_F x_{LK,F}} - \alpha_{LK,HK}\frac{D x_{HK}}{L_F x_{HK,F}}}{(\alpha_{LK,HK})_F - 1}\right]$$

$$\sum \frac{(\alpha_{i,r})_{av}\, z_{i,F}}{(\alpha_{i,r})_{be} - \theta} = 1 - q$$

$$\sum \frac{(\alpha_{i,r})_{av}\, x_{i,D}}{(\alpha_{i,r})_{be} - \theta} = 1 + R_{min}$$

} Solve for θ, $\alpha_{HK} < \theta < \alpha_{LK}$

Distr. of non-key components:

$$\frac{D x_{i,D}}{L_F z_{i,F}} = \left[\frac{(\alpha_{i,HK})_F - 1}{(\alpha_{LK,HK})_F - 1}\right]\left(\frac{D x_{LK,D}}{L_F x_{LK,F}}\right) + \left[\frac{(\alpha_{LK,HK})_F - (\alpha_{i,HK})_F}{(\alpha_{LK,HK})_F - 1}\right]\left(\frac{D x_{HK,D}}{L_F x_{HK,F}}\right)$$

For class I: $0 < \left(\frac{D x_{i,D}}{F x_{i,F}}\right) < 1$ → This tells you if the non-key components are distributing

Drying

$H = \frac{P_A}{P_t - P_A}\left(\frac{18.02}{28.97}\right)$; Absolute humidity ; $RH = (100)\frac{P_{H_2O}}{P^{sat}_{H_2O}}$; Energy needed to achive a certain level of drying

$\frac{H - \tilde{H}_c}{T - T_s} = \frac{-c_p}{\lambda_{H_2O}}$; $c_p = 1.005 + 1.88H$; $\tilde{H}_s = c_p(T - T_{ref}) + H\lambda_{H_2O}$ (absolute H)

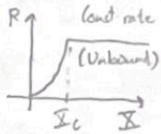
$V_H = [2.83\times10^{-3} + (4.56\times10^{-3})H]\,T$ [=] $\frac{m^3}{kg\ DA}$; $T = 459.67 + F^\circ$; $C = \$/MW$; $\%H = 100H/H_s$

$V_H = [00252 + (0.0405)H]\,T$ [=] $\frac{ft^3}{lb_m\ DA}$

$\bar{X}_t = \frac{W_{total} - W_{dry}}{W_{dry}}$ (free moisture) ; $\frac{d\bar{X}}{dt} = D_t\frac{\partial^2 \bar{X}}{\partial y^2}$; BC: 1) $\bar{X}(y, t=0) = \bar{X}_c$ 2) $\bar{X}(y = y_1, t) = 0$ 3) $\left.\frac{d\bar{X}}{dy}\right|_{y=0} = 0$

$\frac{\bar{X}(t)}{\bar{X}_c} = \frac{8}{\pi^2}\exp\left(-Dt\left(\frac{\pi}{2y_1}\right)^2 t\right)$

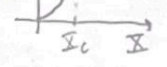
Rate of Drying → $R = -\frac{W_{dry}}{A}\frac{d\bar{X}}{dt} = \frac{q}{\Delta H_{vap}}$; $q = h(T - T_w)$



$\bar{X} = \bar{X}_t - \bar{X}_e$ (Critical free moisture)

$h = 0.0204\,G^{0.8}$ } Parallel flow ; $G = \rho v$; $h = 1.17\,G^{0.37}$ (perp flow)

Time to dry: 1) $t = \frac{W_{dry}}{A\left(\frac{h(T - T_w)}{\lambda \cdot H_2O}\right)}(\bar{X}_c - \bar{X}_f)$ } Const. rate (MT limited) ; MT coefficient



Diffusivity of H_2O in slab

2) $t = \frac{4y_1^2}{\pi^2 D_t}\ln\left(\frac{8\bar{X}_c}{\pi^2\bar{X}_f}\right)$ } Chang. rate (Diff. limited):

Thermodynamics

$EB = \sum_{out}(nh + Q + W_s) - \sum_{in}(ns + \frac{Q}{T_s}) = 0$; T_s = Coolant/heater T ; T_o = surrounding T

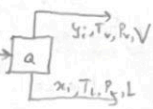
$\eta = \frac{W}{Q_{in}} = 1 - \frac{T_{out}}{T_{in}}$; Entropy balance: $\sum_{out}(ns + \frac{Q}{T_s}) - \sum_{in}(ns + \frac{Q}{T_s}) = \Delta S_{irr}$; $\Delta S_{univ} \geq 0$; LW = Lost Work

$\eta = \frac{W_{min}}{LW + W_{min}}$; Availability balance: $\sum_{in}\left[nb + Q\left(1 - \frac{T_o}{T_s}\right) + W_s\right] - \sum_{out}\left[nb + Q\left(1 - \frac{T_o}{T_s}\right) + W_s\right] = LW = T\Delta S_{irr}$; $W_{min} = \sum_{out} nb - \sum_{in} nb$; $f_{iv} = f_{iL}$

$b = h - T_o S$; $W_{min} + LW$: $\sum_{in} Q_{in}\left(1 - \frac{T_o}{T_{in}}\right) - \sum_{out} Q_{out}\left(1 - \frac{T_o}{T_{out}}\right) + W_{s,in} - W_{s,out}$; $\phi_i y_i P = x_i \gamma_i P_i^{sat}$; $K = \frac{y_i}{x_i} = \frac{y_i \phi_L}{\phi_v} = \frac{\gamma_i P_i^{sat}}{\phi_p} \approx \frac{P_i^{sat}}{P}$

$G = H - TS$; Gibbs: $F = C - P + 2$; System: $\nu = cP + 2$ (Intensive vars T, P) ; $\xi = P + CCP - 1$; $\sum x_i = 1$; $\sum y_i = 1$; $K_i = \frac{y_i}{x_i}$; $F = \nu - \xi$; $\nu = \nu_{int} + (C+1)$; $\xi = \xi_{int} + (C+1)$; $F = \nu - \xi = C + 5$

Flash: $F = C + 5$; $\xi = 3C + 1b$; $1b = \{3T, 3P, F, V, L, Q\}$; $3C = \{z_i, x_i, y_i\}$; $\xi = 2C + 5$ (2 vars undefined, C+3 defined in feed)



$\alpha_{AB} = \frac{K_A}{K_B} = \frac{P_A^{sat}}{P_B^{sat}}$, $\alpha_{AB} > 1$ (Easy separation)

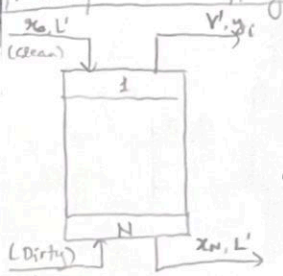
$T_v, \Psi_v \Rightarrow$ Isothermal ; $T_v, V/F = 1 \Rightarrow$ Dew P ; $V/F = 0, P_b \Rightarrow$ Bubble ; $Q = 0, P_v \Rightarrow$ Adiabatic flash ; $V/F = 1, P_v \Rightarrow$ Dew ; $Q, P_v \Rightarrow$ Non-adiab. flash ; $T_L, V/F = 0 \Rightarrow$ Bubble P ; $V/F, P_v \Rightarrow$ % Vap. flash

$P_{bubble}(\Psi = 0) = \sum z_i P_i^{sat}$; $(z_i = x_i)$; $P_i^{sat} = \exp\left(A - \frac{B}{T + C}\right)$; $\sum_{i=1}^{c}\frac{z_i(1 - K_i)}{1 + \Psi(K_i - 1)} = 0$; $\frac{\Psi - 1}{\Psi}x_{i,p} + \frac{1}{\Psi}z = y_A$; $\Psi = \frac{V}{F}$

$P_{dew}(\Psi = 1) = 1/\sum\left(\frac{z_i}{P_i^{sat}}\right)$; $(z_i = y_i)$

Absorption & Stripping

Absorber (g → liq) ; Stripper (liq → gas)



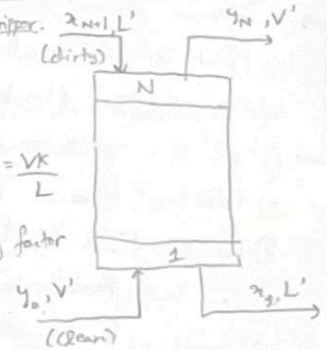
$\bar{X}_n = \frac{x_n}{1 - x_n}$; $\bar{Y}_n = \frac{y_N}{1 - y_N}$; $A = \frac{1}{S}$

$\phi_A = \frac{v_i}{v_{N+1}} = \frac{y_i V'}{y_{N+1} V'}$; $A_e = \frac{L}{VK}$; $\phi_A = \frac{A_e - 1}{A_e^{n+1} - 1}$; opt. $A_e = 1.4$

$\bar{X}_0 L' + \bar{Y}_{n+1} G' = \bar{X}_n L' + \bar{Y}_1 G'$

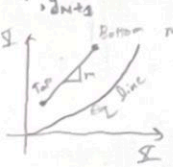
$\left(\frac{L'}{G'}\right)_{min} = \frac{(\bar{Y}_{N+1} - \bar{Y}_1)}{\left[\frac{\bar{Y}_{N+1}}{\bar{Y}_{N+1}(\bar{K}_n - 1)} + \bar{K}_n\right] - \bar{X}_0}$; $\left(\frac{L'}{G'}\right)_{opt} = 1.5\left(\frac{L'}{G'}\right)_{min}$

$\phi_S = \frac{l_2}{l_{N+1}} = \frac{x L'}{x_{N+1} L'}$; $Se = \frac{Se - 1}{Se^{n+1} - 1}$; $Se = \frac{VK}{L}$; Se = stripping factor

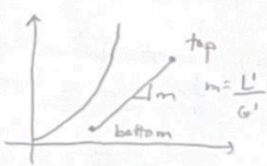


Op. line: $\bar{Y}_{n+1} = \bar{X}_n\left(L'/G'\right) + \bar{Y}_1 - \bar{X}_0\left(L'/G'\right)$ (Absorber)

$\bar{Y}_n = \bar{X}_{n+1}\left(L'/G'\right) + \bar{Y}_0 - \bar{X}_1\left(L'/G'\right)$ (Stripper)



$m = \frac{L'}{G'}$



Packed bed: $h_L = \frac{1}{L}\int_0^L k_x a\,dx$; Length = $H_{OG} \cdot N_{OG}$

H_{OG} [m]

$(y - y^*)$ $V/\bar{K}_y aS$

$(p - p^*)$ $V/\bar{K}_G aPS$

$(\bar{Y} - \bar{Y}^*)$ $V'/\bar{K}_y aS$

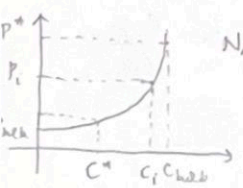
$(x - x^*)$ $L/\bar{K}_x aS$

$(C^* - C)$ $L/[\bar{K}_L a(\rho/M_L)S]$

$(\bar{X} - X)$ $L'/(\bar{K}_x aS)$

$\frac{k_x a}{C_t D_{AB}} = \frac{k_L a}{D_{AS}} = Sh = \alpha\left(\frac{1}{\mu_L}\right)^{1-n}\left(\frac{\mu_L}{\rho D_{AB}}\right)^{0.5}$

$\frac{1}{\bar{K}_G C_g} = \frac{1}{k_g C_g} + \frac{\bar{K}}{k_L C_L}$; $\frac{1}{\bar{K}_L a} = \frac{1}{k_L a} + \frac{1}{\bar{K} k_g a}$



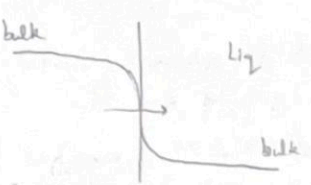
$N_A = k_g(P_{A,i} - P_A) = \bar{K}_G(P_A^* - P_A)$

$= k_L(C_A - C_{A,i}) = \bar{K}_L(C_A - C^*)$

$N_A = k_y(y_{A,i} - y_A) = \bar{K}_y(y_{A}^* - y_A)$

$= k_x(x_A - x_{A,i}) = \bar{K}_x(x_A - x_A^*)$

$a = \frac{area}{volume}$



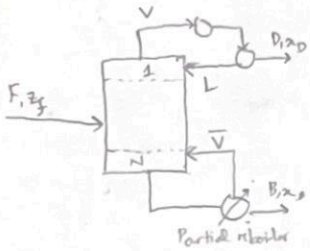
$N_{OG} = \ln\left[\left(\frac{A-1}{A}\right)\left(\frac{y_{in} - K x_{in}}{y_{out} - K x_{out}}\right) + \frac{1}{A}\right] \Big/ \left(\frac{A-1}{A}\right)$

Assumptions:
1) Const L, V
2) No air dissolved in L or L evap. in air
3) Ideal gas for air

Minimum Stripping rate: $V_{min} = \frac{L}{K_{eq}} \cdot$ (fraction of solute stripped)

diffuses

Binary Distillation



$R = \frac{L}{D}$ (Reflux ratio)

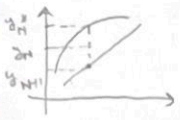
$V_B = \frac{\bar{V}}{B}$ (Reboiler ratio)

Opt. $R = 1.3\,R$

→ Total reboiler/reflux ($R \to \infty \Rightarrow N_{min}$)
↳ Least Capex, highest Opex

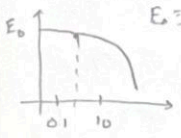


→ Murphree efficiency (How close it reaches eq)



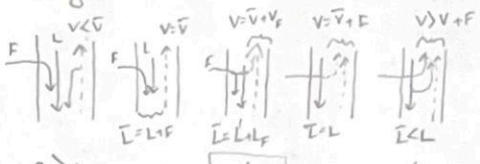
$$E_M = \frac{y_N - y_{N+1}}{y_N^* - y_{N+1}}$$

→ Overall Efficiency



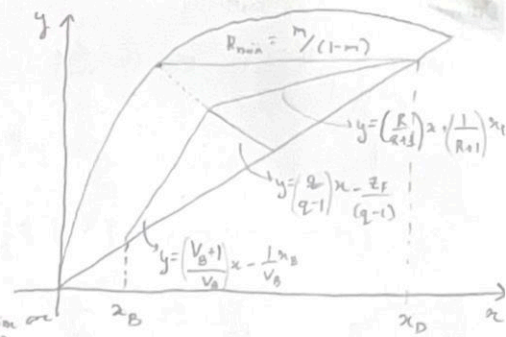
$$E_o = \frac{N_{Theoretical}}{N_{Actual}} = (50.3)(\alpha_{AB}\,\mu_{liq})^{-0.226}$$

Quality:

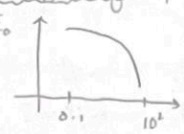


$q>1$	$q=1$	$q=L_F/F$	$q=0$	$q<0$
Subcooled liq	Sat. liq	partially vap.	Sat. vapor	Superheated vapor

McCabe Thiele Assp: No heat loss, Const P, Binary mixt, $c_p\Delta T$ & ΔH_{mix} are negligible, Comp. have equal & const. latent heat



→ Plate efficiency (specific to each stage)



$E_o \propto (K_L M_L \mu_L / \rho_L)^{-1}$

K_L: Eq const. M_L: MW of liq μ_L: Visc. of liq ρ_L: density of liq

E.B on Distillation:

$$Fh_F + Q_R = Dh_D + Bh_B + Q_C + Q_{loss}^{\to 0}$$

Condenser: $Q_C = \underbrace{D(R+1)}_{V}\Delta H^{vap}$

Reboiler: $Q_R = \underbrace{B V_B}_{\bar{V}}\Delta H^{vap}$

$V = \bar{V} + V_F$ (partial vaporized)

$(L+D) = V_B B + V_F$; $Q_R = Q_R - V_F$

$D(R+1) = V_B B + V_F$ $\quad Q_R = Q_R\left(1 - \frac{V_F}{Q_R}\right)$

$= Q_R\left(1 - \frac{V_F}{D(R+1)}\right)$

Multicomponent Distillation

1) Specify splits (LK, HK)
2) Estimate splits of non-key components
3) Determine column pressure & type of condenser (Dew point & Bubble point)
4) Flash the feed @ column pressure (Adiabatic flash)
5) Calculate N_{min} (Fenske)
6) Calculate splits of non-key
7) Calculate $(R_{\infty})_{min}$ (Underwood)
8) Calculate N, R (Gilliland)
9) Calculate Feed location (Kirkbride)
10) Condenser & Reboiler Duties

Class I → narrow-boiling range mixture, low degree of separation

i = LK, j = HK

5) Fenske

$$N_{min} = \frac{\log\left\{\frac{(x_{i,D})}{(x_{i,B})}\frac{(x_{j,B})}{(x_{j,D})}\right\}}{\log \alpha_{ij}} = \frac{\log\left\{\frac{d_i}{d_j}\frac{b_j}{b_i}\right\}}{\log \alpha_m} \; ; \; \alpha_m = \left[(\alpha_{ij})_B (\alpha_{ij})_D\right]^{1/2}$$

6) Non-key dist. (Sanity check)

$$b_i = \frac{f_i}{1 + \left(\frac{d_r}{b_r}\right)(\alpha_{i,r})^{N_{min}}} \; ; \; d_i = \frac{f_i\left(\frac{d_r}{b_r}\right)(\alpha_{i,r})^{N_{min}}}{1 + (d_r/b_r)(\alpha_{i,r})^{N_{min}}}$$

d_r, b_r → HK (reference)

7) Underwood

↳ Class I: $\frac{(L_o)_{min}}{F} = \left(\frac{L_F}{F}\right)\left[\frac{D x_{LK,D}}{L_F x_{LK,F}} - (\alpha_{LK,HK})_F \frac{D x_{HK,D}}{L_F x_{HK,F}}\right] \Big/ \left[(\alpha_{LK,HK})_F - 1\right]$; $(R_{\infty})_{min} = \frac{(L_{\infty})_{min}}{D}$

Non key components

$$0 < \left(\frac{D x_{i,D}}{F x_{i,F}}\right) < 1 \; ; \; \frac{D x_{i,D}}{L_F z_{i,F}} = \left[\frac{(\alpha_{i,HK})_F - 1}{(\alpha_{LK,HK})_F - 1}\right]\left(\frac{D x_{LK,D}}{L_F z_{LK,F}}\right) + \left[\frac{(\alpha_{LK,HK})_F - (\alpha_{i,HK})_F}{(\alpha_{LK,HK}) - 1}\right]\left(\frac{D x_{HK,D}}{L_F z_{HK,F}}\right)$$

↳ Class II (only LK & HK distribute): $\sum_{i=1}^{c}\frac{(\alpha_{i,HK})\,z_{in,F}}{(\alpha_{i,HK}) - \theta} = 1 - q$; $\sum_{i=1}^{c}\frac{(\alpha_{i,HK})\,x_{i,D}}{(\alpha_{i,HK}) - \theta} = 1 + (R_{\infty})_{min} = \frac{(V_{\infty})_{min}}{D}$

$\alpha_{HK} < \theta < \alpha_{LK}$

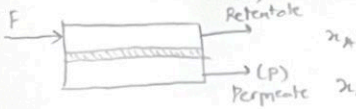
8) Gilliland

$$\frac{N - N_{min}}{(N+1)} = 1 - \exp\left[\frac{(1 + 54.4x)}{(11 + 117.2x)}\frac{(x-1)}{x^{0.5}}\right] \; ; \; x = \frac{R - R_{min}}{(R+1)}$$

9) Kirkbride

$$\frac{N_R}{N_S} = \left[\frac{(z_{HK,F})}{(z_{LK,F})}\left(\frac{x_{LK,B}}{x_{HK,D}}\right)^2 \frac{B}{D}\right]^{0.206} \; ; \; N = N_S + N_R$$

Membrane



$n_F = n_P + n_R$; $\theta = \frac{n_P}{n_F}$; $\frac{n_R}{n_F} = (1-\theta)$

$x_A n_F = y_A n_P + x_A n_R$

$x_A = y_A \theta + x_B (1-\theta)$

$\frac{F_i}{A_i} = N_i = \frac{(P_{mi})}{l_m}$ (driving force) ; $\overline{P}_{mi} = \frac{P_{mi}}{l_m}$ → Permeance ; → Permeability

$\alpha_{AB} = \alpha^*_{AB}\left[\frac{x_A(\alpha_{A,B}-1)+1-r\alpha_{AB}}{x_A(\alpha_{AB}-1)+1-r}\right]$

$\alpha_{AB} = \frac{y_A/x_A}{y_B/x_B}$; $\alpha^*_{AB} = \frac{P_{mA}}{P_{mB}}$

$y_{P_A} = x_{R_A}^{(1/1-\alpha)}\left(\frac{1-\theta}{\theta}\right)\left[(1-x_{R_A})^{\frac{\alpha}{\alpha-1}}\left(\frac{x_{F_A}}{1-x_{F_A}}\right)^{\frac{\alpha}{\alpha-1}} - x_{R_A}^{\frac{\alpha}{\alpha-1}}\right]$

$y_A dn = \frac{P_{m_A} dA_m}{l_m}\left[x_A P_F - y_A P_P\right]$

$A_m = \int_{x_{R_A}}^{x_{F_A}} \frac{l_m y_A dn}{P_{m_A}(x_A P_F - y_A P_P)}$

Porous

Gas: $N_i = \frac{D_{ei} C_m}{P l_m}(P_{i_0} - P_{i_L})$; $C_m = \frac{P}{RT}$; $D_{ei} = \frac{\varepsilon}{\tau}\left[\frac{1}{\frac{1}{D_i} + \frac{1}{D_{K_i}}}\right]$

$P_{m_i} = \frac{D_{ei} C_m}{P}$; $D_{K_i} = \frac{d_p \bar{v}_i}{3}$

Liq: $N_i = \frac{D_{ei}}{l_m}(C_{i_0} - C_{i_L})$; $D_{ei} = \frac{\varepsilon D_i K_{ri}}{\tau}$; $K_{r_i} = \left[1 - \frac{d_{mi}}{d_p}\right]^4$

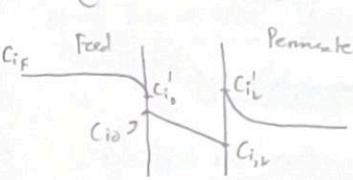
$P_{m_i} = \frac{\varepsilon D_i K_{ri}}{\tau}$ (Permeability)

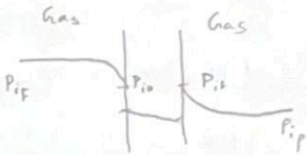
Non-porous

Gas: $N_i = \frac{H_i D_i}{l_m}(P_{i_0} - P_{i_L})$; $H_{i_0} = \frac{C_{i_0}}{P_{i_0}} = \frac{C_{i_L}}{P_{i_L}}$, $P_{m_i} = H_i D_i$

Liq: $N_i = \frac{D_i}{l_m}(C'_{i_0} - C'_{i_L}) = \frac{K_i D_i}{l_m}(C_{i_0} - C_{i_L})$; $K_i = \frac{C_{i_0}}{C'_{i_0}} = \frac{C_{i_L}}{C'_{i_L}}$;

$P_{m_i} = K_i D_i$





Same Line

$F + S = M = E + R$

Same line

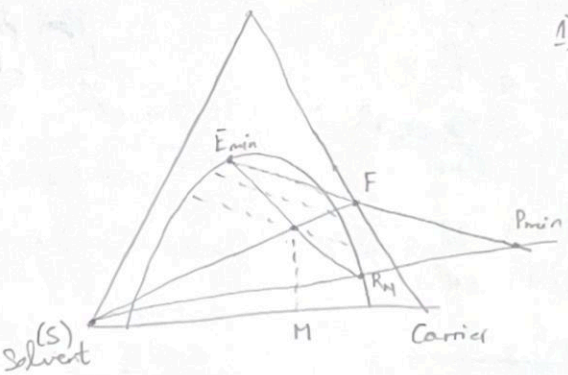
LLE

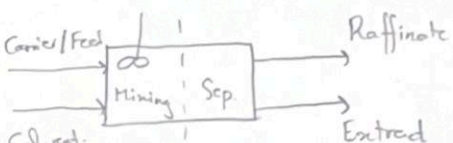
Type II
- Solute & Solvent very miscible.
- requires high flow rate of solvent (S/F)



Type I
- Solute & Solvent more miscible than type II

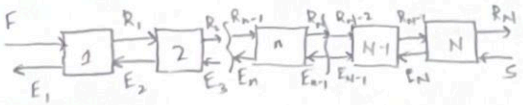


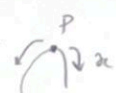


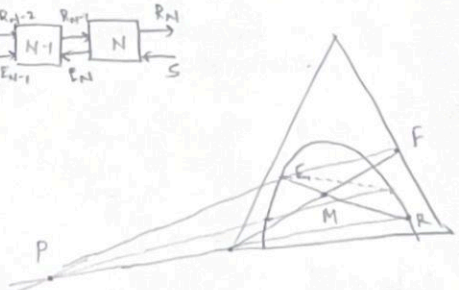


Selectivity = $\frac{K_{solute}}{K_{carrier}} = \frac{(x_{Ac})_E / (x_{Ac})_R}{(x_W)_E / (x_W)_R}$

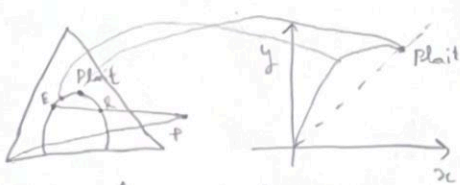
W = carrier
Ac = solute







Converting LLE to McCabe Thiele



y-x from eq tie line data

E-R (points throng P) make up the operating line

1) Finding S_{min} (min. Solvent flow rate)
- Find E using tie line through F to S-R_N line
- Connect E_{min} & R_N
- Find M using graphical method.
- $\frac{S}{M} = \frac{\overline{MF}}{\overline{SF}}$ or $\frac{S}{F} = \frac{\overline{MF}}{\overline{MS}}$

(M = 100 + S)

2) Find stages
- Given S, F → we can find M

$(x_A)_S S + (x_A)_F F = (x_A)_M M$

$(x_A)_M = \infty$
- Find intersection of A and S-F line to find M
- Connect M & R to otherside to find E_1 (This the starting E_1)
- Extend S-R & E-F to meet @ P

Repeat steps until the tie crossed R (given) & then count stages
- Now new R_1 is found using tie line and R is connected to P to find E_2

diffuses

Membrane (Osmosis)

$C = \frac{\text{solute (mol)}}{\text{(Solvent - Solute)}}$ (Solute free basis)

$\pi = P_1 - P_2$; $\pi = C \cdot RT$

$= -\frac{RT}{V_{A_L}} \ln(x_A \gamma_A)$

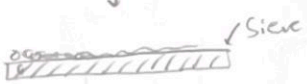
$N_{H_2O} = \frac{P_{m,H_2O}(\Delta P - \Delta \pi)}{l_m}$

S = Salt

→ Salt passage = $SP = \frac{C_{S,P}}{C_{S,F}}$

→ Salt rejection = $SR = 1 - SP$

Random Slit (Membrane)



$J = u = \frac{\Delta P}{\mu(R_m + R_c)}$

- J: Flux
- u: Superficial velocity
- R_m: Resistance from filter
- R_c: Resistance from cake

Total pore vol. / mem volume

↳ for Laminar flow ($R_e < 2100$)

$N = \frac{n \pi \rho D^4}{(128)\mu l_m}(P_0 - P_L) = v_s \varepsilon$; $\varepsilon = \frac{n\pi D^2}{4}$

(n: # pores)

$v = \frac{D^2(P_0 - P_L)}{32 \mu L}$

↳ for hydraulic radius (in industry)

$d_H = \frac{4\varepsilon}{a}$, $a_v = \frac{a}{1-\varepsilon}$

$N = \frac{s\varepsilon^2(P_0 - P_L)}{2(1-\varepsilon)^2 \tau a_v^2 \mu l_m}$; $P_m = \frac{\rho \varepsilon^3}{2(1-\varepsilon)^2 \tau a_v^2 \mu}$

1 barrer = $2.2614 \times 10^{-9} \frac{\text{lbmol}}{\text{hr psia m}}$

$= 10^{-10} \frac{\text{cm}^3(STP)\text{cm}}{\text{cm}^2 \cdot \text{s} \cdot \text{cmHg}}$

For sphere:

$Sh = \frac{k_c D}{D_{AB}} = (4 + 1.21 Pe^{2/3})^{1/2}$ $Pe < 10^4$

Flat plate:

$Sh_L = \frac{k_c L}{D_{AB}} = 0.664 Re^{1/2} Sc^{1/3}$

$Re_0 \to Sc^{1/3}$

gas

diffuses

CBE 150B Cheat Sheet

$C = \frac{P}{MW} = \frac{P}{RT}$

$\bar{V} = \frac{\sum c_i \bar{v}_i}{\sum c_i}$ $\quad v = \frac{\sum \rho_i \bar{v}_i}{\sum \rho_i}$ $\quad N_i \equiv c_i \bar{v}_i$ $\quad n_i \equiv \rho_i \bar{v}_i$ $\quad \bar{N}_i = \bar{J} + c_i \bar{V}$

$\bar{J} = -D_{AB} \bar{\nabla} c_i$ $\quad \bar{j} = -D \bar{\nabla} \rho$ $\quad \bar{n}_i = \bar{j} + c_i \bar{v}$

Balance: $\frac{\partial \rho_i}{\partial t} = -\bar{\nabla} \cdot n_i + r_i$ $\quad \frac{\partial c_i}{\partial t} = -\bar{\nabla} \cdot \bar{N}_i + R_i$

Binary system:

c not const.: $\bar{N}_A = -c D_{AB} \bar{\nabla} y_A + y_A (\bar{N}_A + \bar{N}_B)$ $\quad y_A = \frac{c_A}{c}$

c const.: $\begin{cases} \bar{n}_A = -\rho D_{AB} \bar{\nabla} \omega_A + \omega_A (\bar{n}_A + \bar{n}_B) \\ \bar{N}_A = -D_{AB} \nabla c_A + c_A (x_A v_A + x_B v_B) \end{cases}$ $\quad \omega_A = \frac{\rho_A}{\rho}$

Diffusion: $D_{AB} = -J_A / \nabla c_A$ $\left[\frac{cm^2}{s}\right]$

Gas:
- Ideal → (Molecular) $D_{AA^*} = \frac{1}{3} \lambda u$
  - $\lambda = kT / (\sqrt{2} \pi \sigma_A^2 P)$
  - $u = \left[(8kNT) / \pi M_A \right]^{1/2}$
- Pores → (Knudsen) $K_n = \frac{\lambda}{d_{pore}}$ $\quad K_n > 1$ (Dominates)

Liquid:
- Stokes-Einstein: $D_{AB} = \frac{kT}{6\pi r \mu}$
- Treybal: $\frac{D_{AB} M_B}{T} = \frac{7.4 \times 10^{-8} (\Phi_B M_B)^{1/2}}{V_A^{0.6}}$

$\frac{1}{D_{AE}} = \frac{1}{D_{AB}} + \frac{1}{D_{KA}}$

$D_{KA} = \frac{1}{3} d_{pore} (u) = \frac{d_{pore}}{3} \sqrt{\frac{8RNT}{\pi M_A}} = 4850\, d_{pore} \sqrt{\frac{T}{M_A}}$ (cm, K, g)

$V_A = 0.285\, V_c^{1.048}$ → V_c ⇒ critical volume

2 phases

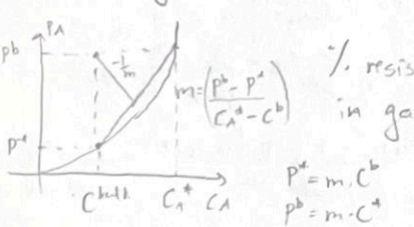
$\text{Flux} = k_G (P^{int} - P^b) = k_L (C^b - C^{int})$

$= \bar{K}_g (P^* - P^b) = \bar{K}_L (C^b - C^*)$

$\frac{1}{\bar{K}_L} = \frac{1}{m k_g} + \frac{1}{k_L}$; $\frac{1}{\bar{K}_g} = \frac{1}{k_g} + \frac{m}{k_L}$

If $m \gg 1$: $\bar{K}_L \sim k_L$ (Liq phase resistance dominates)

$m \ll 1$: $\bar{K}_g \sim k_g$ (gas phase resistance dominates)



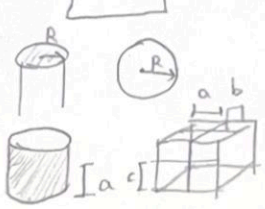
$P^* = m \cdot C^b$ $\quad P^b = m \cdot C^*$

% resistance in gas ⇒ $\frac{1/k_g}{1/\bar{K}_g}$ %

Time dependent Diff in Finite body:

$Y = \frac{C_{As} - C_A}{C_{As} - C_{A0}}$ $\quad X = \frac{D_{AB} t}{x_1^2}$ $\quad m = \frac{D_{AB}}{k_c x_1}$ $\quad n = \frac{x}{x_1}$

One side diffusion in slab: x_1 = L; x_2 = L/2



Convective Mass transfer



- $N_A = k_L (C_{A,s} - C_{A,\infty})$
- $S_c = \frac{\text{momentum diffusivity}}{\text{Mass diffusivity}} = \frac{\mu/\rho}{D_{AB}} = \frac{\nu}{D_{AB}}$ (Thickness of BLS w/ MT BL)
- $Sh = \frac{\text{Convective MT resistance}}{\text{Diffusive MT resistance}} = \frac{k_c L}{D_{AB}}$ (@ interface; or D for sphere)
- $Re_x = \frac{\rho V x}{\mu}$ (inertial / viscous) $\quad Pe_c = \frac{V_\infty L}{D_{AB}}$ (Convective MT / Diff. MT) (in phase)
- $Da = \frac{kL}{D_{AB}}$ (1st order rate / Diff. MT)
- $\frac{\delta}{\delta_c} = Sc^{1/3}$ (BL height; momentum / conc.)
- $\bar{K}_c = \frac{\int_0^L k_{c,x}(x)\, dx}{\int_0^L dx}$
- $Re_x < 2.0 \times 10^5$ (Laminar)

$K_c = \frac{3}{2} \frac{D_{AB}}{\delta_c}$; $\tau_{conv} = \frac{L}{V_\infty}$; $\tau_{diff} = L^2 / D_{AB}$

- Von Karman (Flat plate)

$\frac{C_A - C_{A,s}}{C_{A,\infty} - C_{A,s}} = \frac{3}{2}\left(\frac{y}{\delta_c}\right) - \frac{1}{2}\left(\frac{y}{\delta_c}\right)^3$ $\quad$ Slower τ dominates

$\frac{v_x}{V_\infty} = \frac{3}{2}\left(\frac{y}{\delta_c}\right) - \frac{1}{2}\left(\frac{y}{\delta_c}\right)^2$ $\quad \phi^2 = \frac{k}{D/R^2}$

$\int x^{-1/2}$ $\quad \int x^{-1/5}$

Flat plate: Laminar → $k_c x / D_{AB} = 0.332\, Re_x^{1/2} Sc^{1/3} = Sh$

Turbulent → $k_c x / D_{AB} = 0.0292\, Re_x^{0.8} Sc^{1/3} = Sh$

$j_H = \frac{h}{G c_p} Pr^{2/3} = \frac{Nu_D}{Re_D Pr^{1/3}}$; $j_D = \frac{k_c}{V_\infty} Sc^{2/3} = \frac{Sh}{Re_D Sc^{1/3}}$; $Pr = \frac{c_p \mu}{k}$

(heat transfer coeff; thermal and)

(High volatile) $m \gg 1$: Liq | Gas (low solubility); Gas higher than liquid

$m \ll 1$: Liq | Gas; Liq higher than gas

Liq | Gas: $m \ll 1$, $H < 1$

(Makes sense!)

Diff eq scenarios

$\nabla \cdot N_A + \frac{dc_A}{dt} - R_A = 0$ $\quad N_{B,z} = 0$

(1)



BC:
1) $\left.\frac{\partial c_A}{\partial r}\right|_{r=0} = 0$
2) $C_A(R,t) = C_{eq}^H$

$\frac{1}{r^2}\frac{\partial (r^2 \bar{N}_A)}{\partial r} + \frac{\partial c_A}{\partial t} = 0$

Homogenous, first order (x,y,z)

(2)



$N_{A,z} = -cD_{AB}\frac{dy_A}{dz} + y_A(N_{A,z} + N_{B,z})$

$= \frac{cD_{AB}}{(1-y_A)}\frac{dy_A}{dz} \Rightarrow N_{A,z} = \frac{cD_{AB}}{(1-y_A)}\frac{dy_A}{dz}$

BC's:
1) @ $z=h(t)$, $y_A = y_{A0} = \frac{P_{sat}}{P}$
2) @ $z=z_0$, $y_A = 0$

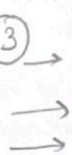
$\int_{h(t)}^{z_2} N_{A,z} = cD_{AB}\ln\left(\frac{1}{1-y_A}\right)$

$N_{A,z} = -\frac{cD_{AB}\ln(1-P_A/P)}{(z_2 - h(t))}$

$\frac{dh}{dt} = \frac{-N_{A,z}MW}{\rho}$

$t_{evap} = \frac{\rho}{MW}\left[\frac{\rho}{RT}D_{AB}\ln\left(1-\frac{P_A^s}{P}\right)\right]^{-1}\left[L(h_f - h_0) - \frac{(h_f^2 - h_0^2)}{2}\right]$

(3)



$v=0$; SS; $C_A = f(z)$; Any BC's

$D_{AB}\frac{d^2C_A}{dz^2} - k_1 C_A = 0$

$C_A(z) = C_1\cosh\left(z\sqrt{\frac{k_1}{D_{AB}}}\right) + C_2\sinh\left(z\sqrt{\frac{k_1}{D_{AB}}}\right)$

(4) C_{As} — Semi infinite Diffusion



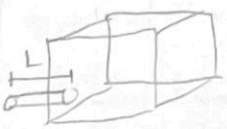
BC's: 1) $\theta(z,t=0) = 0$
2) $\theta(z=0,t) = C_{As} - C_A^b$
3) $\theta(z=\infty,t) = 0$

$\frac{\partial \theta}{\partial t} = D_{AB}\frac{\partial^2 \theta}{\partial z^2}$

$\tau = \frac{\lambda^2}{D_{AB}}$ → penetration depth

$\frac{C_{A,s} - C_A(z,t)}{C_{A,s} - C_{A0}^b} = erf\left(\frac{z}{2\sqrt{D_{AB}t}}\right)$





BC's: 1) $C_A(z=L) = C_{A0}$
2) $C_A(z=0) = 0$

$\nabla \cdot N_A = 0$

$N_A = -D_{AB}\frac{dC_A}{dz} + C_A\bar{V}$

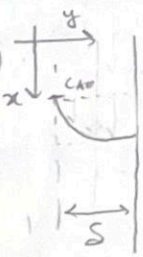
$P_e = \frac{V}{(D/L)}$

A_s

$P_e \to 0$ Diffusion dominates

$P_e \to \infty$ Convection dominates

(5)



$\xi = \delta - y$

$\bar{N}_{A,x} = C_A v_x$
$\bar{N}_{A,y} = -D_{AB}\frac{\partial C_A}{\partial y}$
$\Big\} \; v\frac{\partial C_A}{\partial x} = D_{AB}\frac{\partial^2 C_A}{\partial \xi^2}$

BC's:
1) $C_A(x=0,\xi) = 0$
2) $C_A(x,\xi=0) = C_{A0}$
3) $C_A(x,\xi=\infty) = 0$

$C_A(x,y) = C_{A0}\left(1 - erf\left(\frac{\delta - y}{\sqrt{\frac{4D_{AB}x}{V_{max}}}}\right)\right)$

Cartesian:

$\nabla N = \frac{\partial N}{\partial x}\hat{e}_x + \frac{\partial N}{\partial y}\hat{e}_y + \frac{\partial N}{\partial z}\hat{e}_z$

$\bar{\nabla}\cdot\bar{N} = \frac{\partial N_x}{\partial x} + \frac{\partial N_y}{\partial y} + \frac{\partial N_z}{\partial z}$

Cylindrical:

$\nabla N = \frac{\partial N}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial N}{\partial \theta}\hat{e}_\theta + \frac{\partial N}{\partial z}\hat{e}_z$

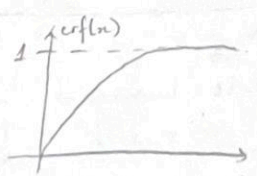
$\bar{\nabla}\cdot\bar{N} = \frac{1}{r}\frac{\partial (rN_r)}{\partial r} + \frac{1}{r}\frac{\partial N_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z}$

Spherical:

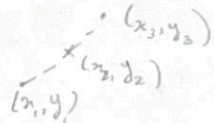
$\nabla N = \frac{\partial N}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial N}{\partial \theta}e_\theta + \frac{1}{r\sin\theta}\frac{\partial N}{\partial \phi}e_\phi$

$\bar{\nabla}\cdot\bar{N} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 N_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(N_\theta \sin\theta) + \frac{1}{r\sin\theta}\frac{\partial N_\phi}{\partial \phi}$

$erf(z) = \frac{2}{\sqrt{\pi}}\int_0^z e^{-z^2}dz$



$\nabla \bar{N}_A = -D_{AB}\nabla c + \bar{v}\nabla c$



$y_2 = \frac{(x_2 - x_1)(y_3 - y_1)}{(x_3 - x_1)} + y_1$

$\frac{d}{dz}\left(erf\left(\frac{z}{2\sqrt{D_{AB}t}}\right)\right) = \frac{2}{\sqrt{\pi}}\exp\left(-\left(\frac{z}{2\sqrt{D_{AB}t}}\right)^2\right)\cdot\frac{1}{2\sqrt{D_{AB}t}}$

$z \to \infty$ or $z \to 0$ → $\exp(-(\cdot)^2) \to 1$

$\frac{d}{dz}(erf(z)) = \frac{1}{\sqrt{\pi D_{AB}t}}$; $\frac{d}{dz}(erf(z)) = \sqrt{\frac{V_{max}}{D_{AB}x\cdot\pi}}$

everything like this (Makes sense!)

CBE 150B Midterm 2 Review

Review:

+ Drying:

$$H = \frac{P_A}{P_{tot} - P_A}\left(\frac{18.02}{28.97}\right)$$

measure of amount of water (mass/mass)

How much the humidity changes the c_p

$$RH = 100 \cdot \frac{P_{H_2O}}{P^{sat}_{H_2O}}$$

Fraction of max water possible

enthalpy of vaporization only for wate +

$$c_p = 1.005 + 1.88H$$

conc. of air

Heat capacity [=] $\frac{kg}{kg\ K}$ dry

Heat capacity increases w/ water present

$$\tilde{H}_s = c_p(T - T_0) + H\lambda_{H_2O}$$

Enthalpy

Always logic check

$$\frac{H - \tilde{H}_s}{T - T_s} = \frac{-c_{p,in}}{\lambda_{H_2O}}$$

Humidity

Humid volume

$$V_H = (2.83 \times 10^{-3} + 4.56 \times 10^{-3} H)T$$ (Kelvin)

Energy needed to achieve a certain level of drying

↓ If you are evap. something then outlet should be cooler

$$X = \frac{W_{tot} - W_{dry}}{W_{dry}}$$

$$R = -\frac{W_{dry}}{A}\frac{dX}{dt}$$

Rate of vaporization

Boundary cond → doesn't have to be sat.

$$q = k(T - T_{sat})$$

$$R = \frac{q}{\Delta H_{vap}}$$

Governing equation: $\frac{\partial X}{\partial t} = D_t \frac{\partial^2 X}{\partial y^2}$

(Diffusion limits evaporation because water has to travel to the surface)



(Finite Slab drying)

Answer:

$$\frac{X(t)}{X_1} = \frac{8}{\pi^2}\exp\left(-Dt\left(\frac{\pi}{2y_1}\right)^2 t\right)$$

1) $X(y, t=0) = X$

2) $X(y = y_1, t) = 0$ (Symmetry)

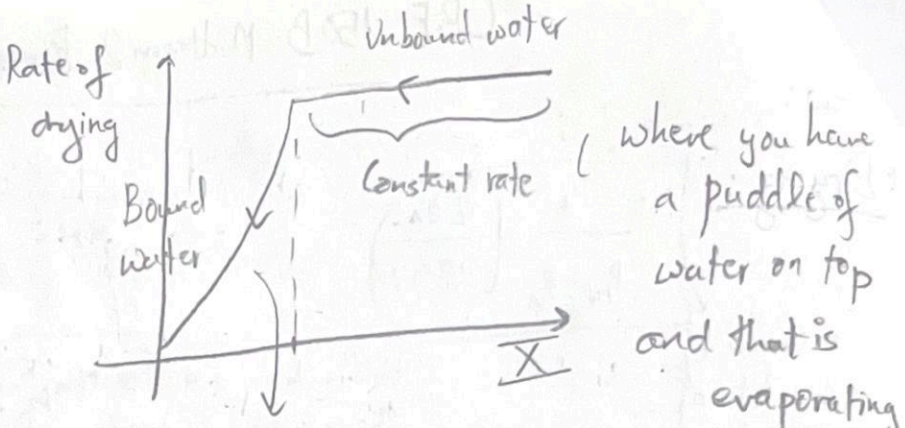
3) $\left.\frac{\partial X}{\partial y}\right|_{y=0} = 0$

everything diffuse (Make sa. 1)

- Time to dry from X_c to X_f

$$t = \frac{W_{dry}}{A\left(\frac{k(T-T_w)}{\lambda_{H_2O}}\right)}(X_c - X_f)$$

- A: Area of plate
- k: Mass transfer coefficient
- λ_{H_2O}: Dependent on mass transfer coefficient



(where you have a puddle of water on top and that is evaporating

This is the rate when water has to diffuse

$$t = \frac{4y_1^2}{\pi^2 D_t} \ln\left(\frac{8}{\pi^2}\frac{X_c}{X_f}\right)$$

- D_t: Diffusivity of water in slab

Bound water is mass transfer limited. because it has to diffuse to the surface to evaporate.

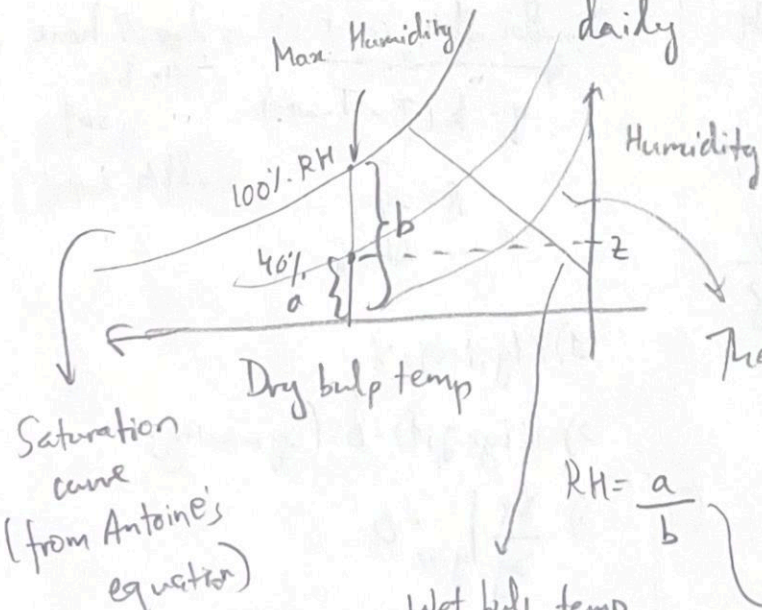
Psychometric Charts.

Wet Bulb: (hypothetical) → Air with 100% humidity (You are surrounding the bulb w/ artificial water)

Dry bulb (physical) → what we measure daily

↓

Used to just get a reference



RH depends on p_{sat}

These are constant RH curves

$RH = \frac{a}{b}$ → This should be the same as the RH curve

H is just z

Eq. T is the wet bulb temp.

ample

If RH↑ → it doesn't mean there is more water in the air.
It rains when H goes down! (It will only go down when it hits 100% RH)

If wet bulb = 73°F
dry " = 80°F
} Read the intersection and this gives the the RH (when RH is not given)

★ At H=0, the wet bulb is lower than dry bulb as the the water from the wet cotton evaporates instantly as there is no water in the air

Thermodynamics

$\eta_{Carnot} = \frac{W}{Q_{in}} = 1 - \frac{T_{out}}{T_{in}}$; $0 = \sum_{out-in} (F_i \tilde{H}_i + Q + W)$; $\Delta S_{irr} = \sum_{out-in} \left(F_i \tilde{S}_i + \frac{Q}{T_{out-in}} \right)$

Availability (Energy): How much energy is available for (without including the heat lost to the env.) ✓

$\tilde{B} = \tilde{H} - T_0 \tilde{S}$ ($G = H - TS$, ↑ T of system)

T_0 is temp of surrounding (outside temp) → Therefore you subtract out the $T_0 S$ (energy) that is lost to the surr.

Gibbs includes the heat loss

$W_{min} = \sum_{out} F_i \tilde{B}_i - \sum_{in} F_i \tilde{B}_i$

$W_{min} + W_{lost} = \sum_{in} Q_{in} \left(1 - \frac{T_0}{T_{in}} \right) + W_{s,in} - \sum Q_{our} \left(1 - \frac{T_0}{T_{out}} \right) + W_{s,out}$

Phase

fugacity coefficient

$K_i = \frac{y_i}{x_i} = \frac{y_{iL} \phi_L}{\phi_{iV}} = \frac{y_{iL} P_i^{sat}}{\phi_{iV} P} \approx \frac{P_i^{sat}}{P}$

partition coefficient

Relative volatility: $\alpha_{AB} = \frac{K_A}{K_B} = \frac{P_A^{sat}}{P_B^{sat}}$

Flash: $\frac{\psi-1}{\psi}x_A + \frac{1}{\psi}z_A = y_A$ $\quad \psi = V/F$

$$\sum_{i=1}^{c} \frac{z_i(1-K_i)}{1+\psi(K_i-1)} = 0$$

Rachford-Rice equation (Solve for V/F)

OP OG

If $\psi = 0$, $V = 0$ (no vapor)
↳ Therefore bubble point

If $\psi = 1$, $L = 0$ (no liquid)
↳ Dew point

$$P_{bubble} = \sum(z_i P_i^{sat})$$

$$P_{dew} = \frac{1}{\sum\left(\frac{z_i}{P_i^{sat}}\right)}$$

This is useful in case L_0 & L_n are not the same (as an absorber would gain mass)

$$\overline{X}_0 L' + \overline{Y}_{n+1} G' = \overline{X}_n L' + \overline{Y}_1 G'$$

Therefore we do it only in terms of the solvent (taking out the solutes from the equation)

Absorber: $\overline{Y}_{n+1} = \overline{X}_n\left(\frac{L'}{G'}\right) + \overline{Y}_1 - \overline{X}_0\left(\frac{L'}{G'}\right)$

Stripp: $\overline{Y}_n = \overline{X}_{n+1}\left(\frac{L'}{G'}\right) + \overline{Y}_0 - \overline{X}_1\left(\frac{L'}{G'}\right)$

$$L'_{min} = \frac{G'(\overline{Y}_{n+1} - \overline{Y}_1)}{\left[\frac{\overline{Y}_{n+1}}{\overline{Y}_{n+1}(K_n - 1)} + K_n\right] - \overline{X}_0}$$

Absorption / Stripping:

Absorber (gas to liquid)

Stripper (liq to gas)

Count from the stream that is clean
↳ We label a stream based on whatever stage it leaves.

Solut free (useful for ternary system that converts it to a binary system)

$$\overline{X}_n = \frac{x_n}{1-x_n} \qquad \overline{Y}_n = \frac{y_n}{1-y_n}$$

$$\phi_A = \frac{V_1}{V_{n+1}} \qquad \phi_S = \frac{l_1}{l_{n+1}}$$

l, v: Component that you are trying to remove

$\phi_A = 1$ (worst separation)

$$A_e = \frac{L}{VK} \; ; \; S_e = \frac{VK}{L}$$

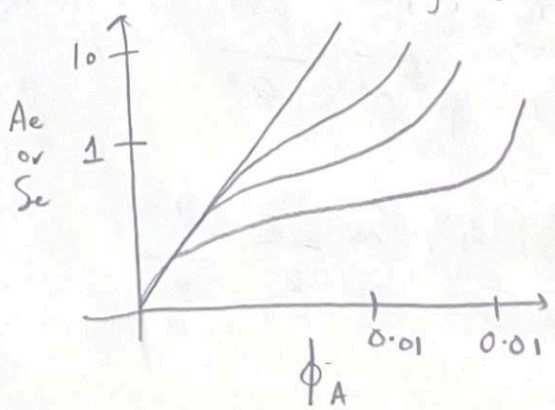
$$\phi_A = \frac{A_e - 1}{A_e^{n+1} - 1} \; ; \; \phi_S = \frac{S_e - 1}{S_e^{n+1} - 1}$$

fraction not removed

$\frac{\text{clean}}{\text{dirty}}$

Round up for n always

n = no. of stages (Can't solve these in class unless known n)

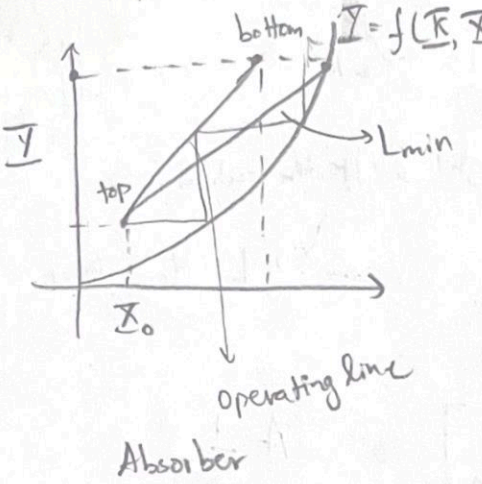


Kremser plot

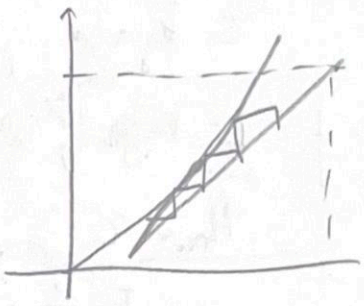
everything diffuses

(Makes sense!)

ng solute free method, we can construct $\underline{Y}$ vs. $\underline{X}$ graphs.



Absorber



Stripper

everything

Packed Column separator: → for continuous separation (no stages)



H_{OG} , N_{OG}

↑ Height of effective plates

$S = \bar{K}_y a (y_1 - y^*) = \bar{K}_x a (x^* - x_b)$

L = length of column = height × effective number

$H_{OG} = \frac{G'}{\bar{K}_g a P S^2}$; $A = \frac{L'}{k_{eq} G'}$ → $L = H_{OG} \cdot N_{OG}$

$$N_{OG} = \frac{\ln\left[\left(\frac{A-1}{A}\right)\left(\frac{y_{in} - k_{eq} x_{in}}{y_{out} - k_{eq} x_{in}}\right) + \frac{1}{A}\right]}{\left(\frac{A-1}{A}\right)} \qquad A = \frac{1}{S}$$

Mass Transport: $\frac{k_x a}{C_{tot} D_{AB}} = \frac{k_L a}{D_{AB}} = Sh = \alpha \left(\frac{L}{\mu_L}\right)^{1-n} \left(\frac{\mu_L}{\rho D_{AB}}\right)^{0.5}$

local MT coefs → $k_x a$; Length of column → L

average MT coeff. $k_L = \frac{1}{L}\int_0^L k_x \, da$

Binary Distillation

$R = \frac{L}{D}$; $V_B = \frac{\bar{V}}{B}$; $q = \frac{\bar{L} - L}{F}$ → how vaporized the feed is

$R \to \infty$ (total reflux)

Heating & Mixing Shit

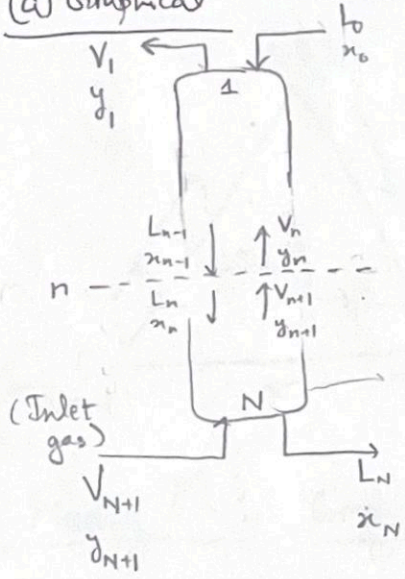
* Absorption / stripping:

Two ways to solve this: (a) Graphical: McCabe-Thiele (any VLE)

(b) Analytical: Kremser (assume dilute solution VLE)

↳ Henry's Law & constant L/V

(a) Graphical



+ Overall material balance (@ the ends):

$$y_{N+1}V_{N+1} + x_0L_0 = y_1V_1 + x_NL_N$$

+ Balance from 1 → n stages

$$y_{n+1}V_{n+1} + x_0L_0 = y_1V_1 + x_nL_n$$

↓ rearranging this equation,

$$y_{n+1} = \frac{y_1V_1 + x_nL_n - x_0L_0}{V_{n+1}}$$

$$= y_1\frac{V/}{V_{n+1}} + x_n\frac{L_n}{V_{n+1}} - x_0\frac{L_0}{V_{n+1}}$$

$n \neq N$ (total no. of stages)

↑ Stages from the top

VLE & $\frac{L}{V}$ is constant (Ideal stages)

$$y_N = \left(\frac{L}{V}\right)x_N + \left(y_1 - \frac{L}{V}x_0\right) \quad \Longleftrightarrow \quad y_{n+1} = \underbrace{\left(\frac{L_n}{V_{n+1}}\right)}_{\text{gradient}}x_n + \underbrace{\left(y_1 - \frac{L_0}{V_{n+1}}x_0\right)}_{\text{intercept}} \rightarrow$$

This becomes the operating line.

Small technicality:

We define $y_A = \frac{\text{moles of solute A (in gas)}}{\text{total moles of gas mixture}}$

However, the gas is in two different mixtures

$\therefore y_A = \frac{\text{A in gas}}{[\text{moles of carrier gas} + \text{mole of solute B}]}$

$$\therefore y_1 = \overline{Y}_1 = \frac{y_A}{1-y_A}$$

Similarly, $\overline{X}_1 = \frac{y_B}{1-y_B}$

Therefore the final equations we get:

$$\overline{Y}_{N+1} = \left(\frac{L}{V}\right)\overline{X}_N + \left(\overline{Y}_1 - \frac{L}{V}\overline{X}_0\right)$$

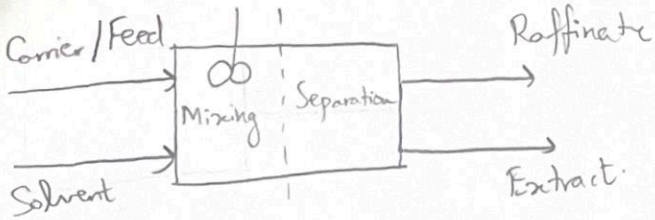
Steps:

1) Draw eq line from Henry's Law

2) Draw operating line

($\overline{Y}_{N+1}$, $\overline{X}_1$, $\frac{L}{V}$, $\overline{X}_N$ are known)

Liquid - Liquid Extraction

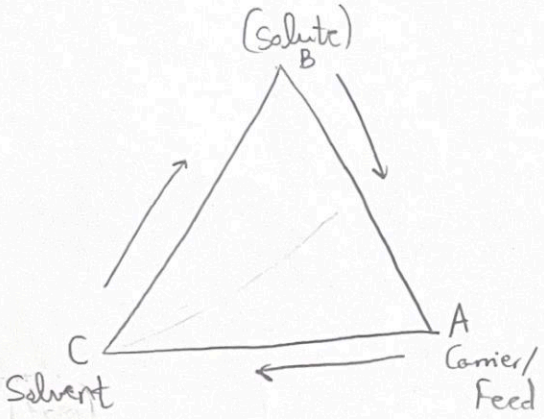


→ The 'Carrier' carries the 'solute'

↔ This is what you need to extract (the solute from the carrier)

The Raffinate carries the unwanted carrier/feed (and some small amounts of remaining solute)

The Extract carries the desired solute and some amount carrier that might have slightly dissolved.



Finding M: Graphically: $\underbrace{F+S}_{\text{lie on same line}} = \overbrace{M}^{\text{lie on same line}} = E+R$ (Shits goin = shits coming out)

Gaurav: The legend

$$\vec{\nabla} \cdot \vec{N}_A + \frac{\partial C_A}{\partial t} - R_A = \emptyset$$ → Make sure to use this correctly according to diff coordinates.

(divergence of a vector is a scalar)

Fick's Law (C is constant)

$$\vec{J}_A = -D_{AB} \vec{\nabla} C_A$$

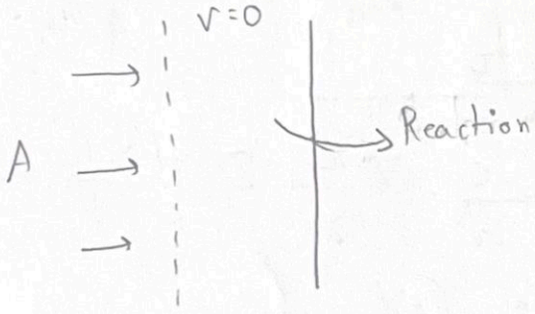
$$\vec{N}_A = \vec{J}_A + \underbrace{C_i V}_{\text{Convection}}$$

(2nd ODE - linear)

Different cases:

① First order Homogenous Rxn: $D_{AB} \frac{d^2 C_A}{dz^2} - k_1 C_A = 0$ $C_A = f(z)$ + (Cartesian) + SS + Any BC's

$$C_A(z) = C_1 \cosh\left(z\sqrt{\frac{k_1}{D_{AB}}}\right) + C_2 \sinh\left(z\sqrt{\frac{k_1}{D_{AB}}}\right)$$



$C_{A_0} = C_{A_1}^{bulk}$

② Semi-infinite Diffusion:

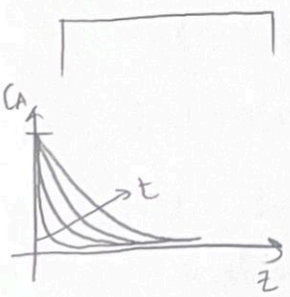
$$\frac{\partial \theta}{\partial t} = D_{AB} \frac{\partial^2 \theta}{\partial z^2}$$ (Partial diff. eq)

$\theta = C_A - C_{A_0}$ $\theta(z, t=0) = 0$

$\theta(z=0, t) = C_{A_s} - C_{A_0}$

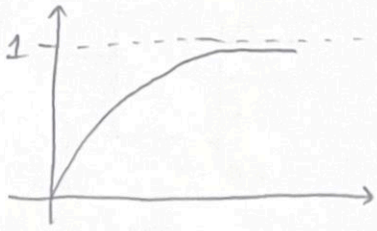
Semi Infinite → $\theta(z=\infty, t) = 0$

A



Solution:

$$\frac{C_{A,s} - C_A(z,t)}{C_{A,s} - C_{A_0}} = \text{erf}\left(\frac{z}{2\sqrt{D_{AB} t}}\right)$$



Check

. When erf(z) = 1

z is very big,

$C_{A,s} = C_A(z,t) = C_{A,s} - C_{A_0}$

$C_A(z,t) = C_{A_0}$

(Makes sense!)

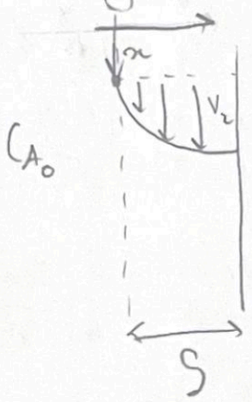
$$\text{erf}(x) = \int_0^{\infty} e^{-x^2} \cdot dx$$

. when $t \to \infty$, erf(0) = 0

$C_{A,s} = C_A(z,t)$

↓ everything diffuses

③ Falling film



$\vec{N}_{A,x} = C_A v_z$ (Ignore diffusion)

$\vec{N}_{A,y} = -D_{AB}\frac{\partial C_A}{\partial y}$ (Only diffusion)

$$v\frac{\partial C_A}{\partial x} = D_{AB}\frac{\partial^2 C_A}{\partial \xi^2}$$

$$\nabla \cdot N_A = \frac{dN_{A,x}}{dx} + \frac{dN_{A,y}}{dy}$$

BC's: $C_A(x=0, \xi) = 0$

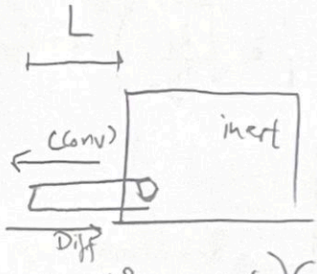
$C_A(x, \xi=0) = C_{A,0}$

* $C_A(x, \xi=\infty) = 0$

$$C_A(x,y) = C_{A0}\left(1 - \operatorname{erf}\left(\frac{y-S}{\sqrt{\frac{4D_{AB}z}{v_x}}}\right)\right)$$

It appears as a semi-infinite as the convection in x is much faster than the diffusion in y

The δ is infinite in a way (x dimension the same as t)



$A = O_2$

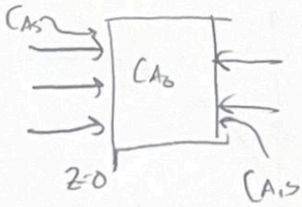
$$D_{AB}\frac{\partial^2 C}{\partial z^2} - v\frac{\partial C}{\partial z} = 0$$ (Diff = Convection)

④ Gloverbar

$$N_A = -D_{AB}\frac{dC_A}{dz} + C_A v$$

Solution: $$C_A = C_{A0}\left(\frac{\exp\left(vz/D_{AB}\right) - 1}{\exp\left(vL/D_{AB}\right) - 1}\right)$$

⑤ Finite Slab



$$\frac{\partial C_A}{\partial t} = D_{AB}\frac{\partial^2 C_A}{\partial z^2}$$

$$Y = \frac{C_{A,S} - C_A}{C_{A,S} - C_{A0}}$$

$Y(z, t=0) = 1$

$Y(z=0, t) = 0$

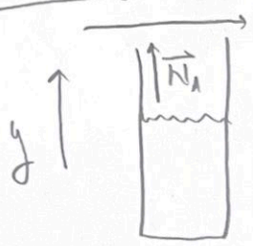
* $Y(z=L, t) = 0$

$$Y(z,t) = \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\left(\frac{n\pi z}{L}\right)\exp\left(-\left(\frac{n\pi}{2}\right)^2\left(\frac{4D_{AB}t}{L^2}\right)\right)$$

↳ Sturm-Liouville Theorem



Molecular diffusion:



Constant flux: $N_A = -D_{AB} \frac{\partial C_A}{\partial y}$

Linear:

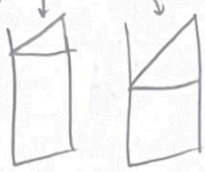


PSS → liquid height does not change

↓

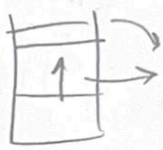
Conc. only depends on the VLE partial pressure

Conc. profile



Higher flux

But we are looking at snapshot of flux in space



Both fluxes at these height are the same.

↓

That's why flux is constant (although changing w/ time)

Non-Dimensional no.:

ν = kinematic viscosity

① $Re = \frac{\rho v D}{\mu} = \frac{vD}{\nu} = \frac{\text{inertial}}{\text{viscous}}$

② $Sc = Pr_m = \frac{\nu}{D_{AB}} = \frac{\text{diff momentum transfer}}{\text{diff mass transfer}}$

③ $Le = \frac{\lambda}{\rho C_p D_{AB}} = \frac{\text{thermal diffusion}}{\text{mass diffusion}}$ (λ: Thermal conductivity)

④ $Sh = Nu_m = \frac{k_c L}{D_{AB}} = \frac{\text{convective Mass T}}{\text{diffusive MT}}$ (at an interface)

⑤ $Pe_c = \frac{v_\infty L}{D_{AB}} = Re\ Pr = \frac{\text{Convective Mass T}}{\text{Diffusive Mass T}}$ (in same phase)

⑥ $Da = \frac{kL}{D_{AB}} = \frac{\text{1st order rate}}{\text{Diffusive MT}}$

⑦ $K_n = \frac{\lambda}{d_{pore}} = \frac{\text{diffusion (normal)}}{\text{knudson Diffusion}}$

High k_L, low resistance (things can move faster)

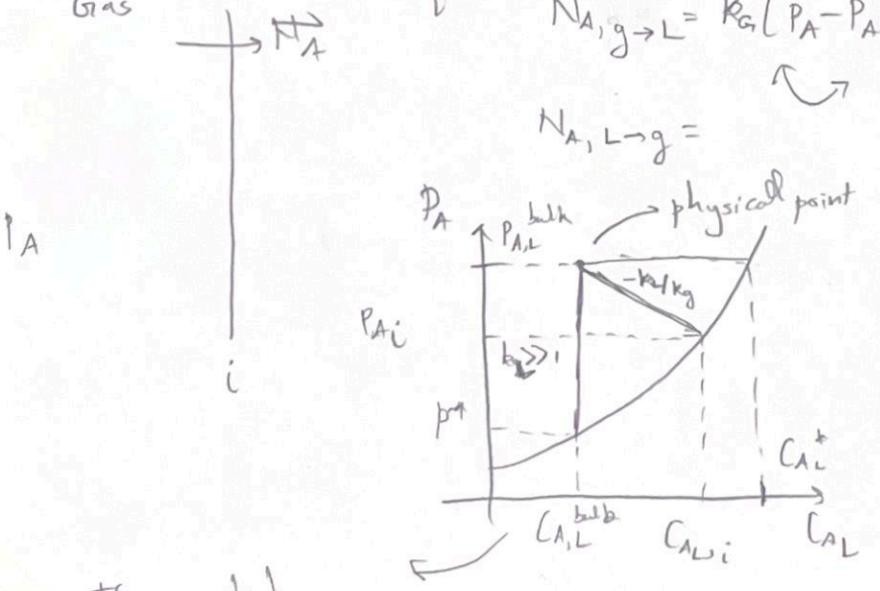
↑ (How fast can stuff move in liquid)

Liquid transfer coefficient

↓ resistance ($\frac{1}{k_A}, \frac{1}{k_a}$)

2 phases

Gas | Liquid



$N_{A,g\to L} = k_G(P_A - P_{A,i}) = k_L(C_{A,L,i} - C_{A,L})$

$N_{A,L\to g} =$

$N_{A,G\to L} = \overline{K}_G(P_A - P_A^*)$

$= \overline{K}_L(C_A^* - C_{A,L})$

$\frac{1}{\overline{K}_G} = \frac{1}{k_G} + \frac{H}{k_L}$

$\frac{1}{\overline{K}_L} = \frac{1}{H k_G} + \frac{1}{k_L}$

If you take bulk conc.
↳ what will be the pressure @ equil

Chem 150B

$\bar{V}$ → Molar average velocity = $\frac{\sum(c_i \bar{v}_i)}{\sum c_i}$

$\bar{v}_i$ → absolute velocity

$\bar{v}$ → Mass average velocity = $\frac{\sum(\rho_i \bar{v}_i)}{\sum \rho_i}$

Molar flux ↓ $N_i = c_i \bar{v}_i$; Molar Diffusion ↓ $\bar{J} = c_i \bar{v}_{D_i} = -D_{AB} \bar{\nabla} c_i$; Mass Diffusion ↓ $j = -D_\rho \bar{\nabla} \omega_i$ $\left(\omega = \frac{\rho_i}{\rho}\right)$

$n_i = \rho_i \bar{v}_i$; $\bar{J}_i = -D_{AB} c \nabla y_i$ $\left(y_i = \frac{c_i}{c}\right)$

↑ Mass flux

$$\boxed{\bar{N}_i = \bar{J}_i + c_i \bar{V}}$$

For a binary mixture:

(i) ⋯ $\bar{N}_A = -c D_{AB} \bar{\nabla} y_A + y_A (\bar{N}_A + \bar{N}_B)$

$\bar{N}_B = -c D_{BA} \bar{\nabla} y_B + y_B (\bar{N}_A + \bar{N}_B)$

$\bar{n}_A = \underbrace{-\rho D_{AB} \bar{\nabla} \omega_A}_{\text{Diffusive term}} + \underbrace{\omega_A (\bar{n}_A + \bar{n}_B)}_{\text{Convective term}}$

Mass Balance:

We know $\frac{\partial \rho}{\partial t} + (\nabla \cdot \rho \bar{v}) = 0$

$$\therefore \boxed{\frac{\partial \rho_i}{\partial t} + \bar{\nabla} \cdot n_i = r_i} \quad ; \quad \boxed{\frac{\partial c_i}{\partial t} + \bar{\nabla} \cdot \bar{N}_i = R_i} \quad \text{(ii)}$$

We can combine equations for a binary system to get: (i) & (ii)

$$\underbrace{-D_{AB} \nabla^2 c_A}_{\text{Diffusion}} + \underbrace{\bar{V} \cdot \nabla c_A}_{\text{Convection}} + \underbrace{\partial c_A / \partial t}_{\text{Non-Steady}} - \underbrace{R_A}_{\text{Chemical reaction}} = 0$$

Hella Diffusion:

$$D_{AB} = \frac{-J_A}{\nabla C_A} \quad \left[\frac{cm^2}{s}\right]$$

Gas $\approx 10^{-2} - 10^{-1}$
Liq $\approx 10^{-4} - 10^{-5}$
Sol $\approx 10^{-10} - 10^{-6}$

There are various models describing the diffusion of various species in various phases.

$D_{AB} \propto T, P^{-1}$, composition

Gases

(Kinetic Theory)

- Ideal gas

$$D_{AM} = \frac{1}{3}\lambda u$$

$$\lambda = \frac{kT}{\sqrt{2}\pi\sigma_A^2 P}$$

$$u = \sqrt{\frac{8RNT}{\pi M_A}}$$

- Hicchfelder

non-reacting, low P, non-polar

$$D_{AB} = \frac{0.001858\, T^{3/2}\left(\frac{1}{M_A} + \frac{1}{M_B}\right)^{1/2}}{P\sigma_{AB}^2 \Omega_D}$$

- Fuller

when σ & $\Delta\varepsilon$ are not available

$$D_{AB} = \frac{0.001\, T^{1.75}\left(\frac{1}{M_A} + \frac{1}{M_B}\right)^{1/2}}{P\left[(\Sigma v_i)_A^{1/3} + (\Sigma v_i)_B^{1/3}\right]^2}$$

Liquid

→ Schiebel: $\frac{D_{AB}\mu_B}{T} = \frac{K}{V_A^{1/3}}$ (Analytical)

- Stokes Einstein

$$D_{AB} = \frac{kT}{6\pi r_A \mu_B}$$

- Wilke & Chang

$$\frac{D_{AB}\mu_B}{T} = \frac{7.4\times10^{-8}\left(\phi_B M_B\right)^{1/2}}{V_A^{0.6}}$$

$V_A = 0.285\, V_c^{1.048}$

(If V_A not available)

Pore Diffusivity

Diffusion of gas molecules through small pores

$$K_n = \frac{\lambda}{d_{pore}}$$

$K_n \propto P$

(As more collisions with porewalls)

$0.1 < K_n < 1$ (Moderate)

$K_n > 1$ (Important)

$K_n > 10$ (Dominate)

(K_n does not work for liquids as λ is very small)

$$\frac{1}{D_{AE}} = \frac{1}{D_{AB}} + \frac{1}{D_{KA}}$$

'Series' competition of Knudsen & molecular (in a dilute mixture)

⇓

Straight, cylindrical pores

D_{AE}: effective diffusion (within the pore)

D_{AB}: Molecular Diffusion